Assessment for Senior Engineer - Technology Quality Assurance

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Target Application: https://restful-booker.herokuapp.com

Core Competency: Managing complex transactional states, API chaining under concurrency, idempotency validation, and handling shared-environment flakiness.

Background & Challenge

restful-booker is a shared public sandbox. In a real enterprise microservices environment, tests frequently fail due to shared state pollution, race conditions, and transient auth issues.

Design an advanced, resilient automated test pipeline that executes a complex booking lifecycle while gracefully managing concurrency and data constraints.

Detailed Requirements

1. Advanced Authentication & Token Lifecycle Management

• **Dynamic Re-auth Simulation:** Do not rely on a single static token. Implement an auth manager that checks token validity, handles token expiration, and dynamically triggers re-authentication if an unauthorized (401/403) response is intercepted mid-suite.
• **Concurrent Token Generation:** Ensure that if multiple test threads execute parallel setup hooks, the token generation endpoint does not race or lock up.

2. Complex Business Logic & Date-Range Validation Graph

• **Overlapping Bookings (Negative State Validation):** Programmatically calculate check-in and check-out dates. Attempt to create a duplicate booking for the exact same room/date range using a second user context, and assert that the system rejects it with proper validation codes (e.g., 400 or 409 conflict, or appropriate sandbox behavior).
• **Deep Schema Contract & Invariant Validation:** Validate that price calculations, deposit paid flags, and nested JSON structures (bookingdates object containing checkin and checkout) strictly conform to date-time formats and logical constraints (checkout > checkin).

3. Idempotency & Safe Retry Mechanics

• **Idempotent Updates:** Write a test verifying that sending the exact same PUT request payload three times consecutively yields an identical, successful state without side effects or escalating resource versions.
• **Partial Update Drift (PATCH):** Send a PATCH request updating only a single field (e.g., totalprice), and verify that all other unreferenced fields (such as firstname, lastname, and bookingdates) remain entirely untouched and uncorrupted.

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4. Concurrency & Race Condition Contention (The "Senior" Test)

• **Simultaneous Modification:** Write a concurrent test script (using async/threading) where two distinct threads attempt to modify the same booking ID with conflicting data at the exact same millisecond.
• **State Reconciliation:** Assert how the system handles the collision (last-write-wins vs. optimistic locking failure) and verify that the final database state is deterministic and free of data corruption or partial payload merges.

5. Fault-Tolerant Setup, Teardown, and Environment Isolation

• **Shared Sandbox Hygiene:** Because the target app is a public sandbox where data can be modified or deleted by other users globally, your framework must include resilient failure recovery. If a GET or DELETE returns a 404 because another user purged the record, the test framework must handle it gracefully (e.g., auto-recreating the prerequisite state) rather than failing the entire pipeline due to dirty data.

Assessment Guidelines & AI Usage Policy

Note on Agentic AI Assistance:

Candidates are permitted to use Agentic AI tools and

coding assistants for productivity, code generation, and framework building during the

offline/preparation phase. However, please note that the live evaluation, code walkthrough,

and technical defense phase will be conducted entirely without the assistance of Agentic AI.

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